

Acronyms

LID — Language Identification

Detecting *which language* is being spoken in an audio clip. For example, automatically determining that a caller is speaking French rather than English, before routing their call to the right model. Output is typically a probability distribution over a set of supported languages.

AID — Accent Identification

Detecting *which accent or dialect variant* of a language is being spoken. For example, distinguishing Canadian French from African French, or Indian-accented English from Australian English. LID tells you the language; AID tells you the regional or cultural variety within that language.

SER — Speech Emotion Recognition

Detecting the *emotional state* of a speaker from their voice signal — not from the words they say, but from how they say them.

ASR — Automatic Speech Recognition

Converting spoken audio to written text — what most people call "speech to text" or STT. Whisper and SenseVoice both do this. In your pipeline this produces the transcript that feeds your text-based keyword classifier (CamemBERT for emergency keywords).

AED — Audio Event Detection

Identifying *non-speech sounds* in an audio signal.

VAD — Voice Activity Detection

Detecting *when someone is actually speaking* versus silence or background noise in an audio stream. Used as a preprocessing step to segment a continuous call recording into utterance-level chunks before feeding them to LID/SER/ASR models.

WER — Word Error Rate

The standard accuracy metric for ASR. Measures the percentage of words in a transcript that are wrong (substituted, deleted, or inserted) compared to the correct transcript. Lower is better. A WER of 5% means 5 words in every 100 are incorrect.

CER — Character Error Rate

Same concept as WER but measured at the character level instead of the word level. Used for languages like Chinese or Japanese where word boundaries are ambiguous.

RTF / RTFx — Real Time Factor

A latency metric for speech models. RTF of 1.0 means the model takes exactly as long to process audio as the audio itself lasts. RTFx is the inverse — an RTFx of 10× means the model processes 10 seconds of audio in 1 second. Higher RTFx is faster.

Publicly Available Datasets by Feature

LID — Language Identification

Dataset	Size	Languages	License	Access
VoxLingua107	6,628 hrs	107 languages	CC-BY 4.0	SEACrowd/voxlangua on HuggingFace / OpenSLR
FLEURS	~12 hrs/lang	102 languages	CC-BY 4.0	google/fleurs on HuggingFace
Mozilla Common Voice 18	EN ~3,000 hrs, FR ~900 hrs	100+ languages	CC-0	mozilla-foundation/common_voice_18_0 on HuggingFace/commo nvoice.mozilla.org
VoxPopuli	400K hrs unlabelled, 1.8K hrs labelled	23 European languages	CC-0	facebook/voxpathuli on HuggingFace
MLS (Multilingual LibriSpeech)	44.5K hrs EN + 6K hrs others	EN, DE, FR, ES, IT, PT, NL, PL	CC-BY 4.0	facebook/multilingual _librispeech on HuggingFace

AID — Accent Identification

Dataset	Size	Accents covered	License	Access
CommonAccent (via Common Voice)	~1,200 hrs EN	16 EN accents incl. African, Canadian, Indian, Australian, UK variants	CC-BY 4.0	DTU54DL/common-accent on HuggingFace
CommonAccent DE	Common Voice 11.0 subset	4 German accents (Germany, Austria, Switzerland, Liechtenstein)	CC-BY 4.0	Derived from mozilla-foundation/common_voice_11_0
CommonAccent ES	Common Voice 11.0 subset	6 Spanish accents	CC-BY 4.0	Derived from mozilla-foundation/common_voice_11_0
CommonAccent IT	Common Voice 11.0 subset	5 Italian accents	CC-BY 4.0	Derived from mozilla-foundation/common_voice_11_0
African Accented French (OpenSLR 57)	22 hrs	Cameroonian/Chad, Congo, Niger/Gabon FR	CC-BY 4.0	openslr.org/57 / gigant/african_accented_french on HuggingFace
AESRC-2020	~160 hrs	10 EN accents: Chinese, American, British, Korean, Japanese, Russian, Indian, Portuguese, Spanish, Canadian	Research	openslr.org / contact authors

Afrispeech-200	200+ hrs	120 African accents (pan-African English), 2,463 speakers, 13 countries	CC BY-NC-SA 4.0	tobiolatunji/afriSpeech-200 on HuggingFace
L2-ARCTIC	~27 hrs	Non-native EN: Hindi, Korean, Mandarin, Spanish, Arabic, Vietnamese	CC-BY 4.0	KoelLabs/L2Arctic on HuggingFace

SER — Speech Emotion Recognition

Dataset	Size	Language	Emotions	License	Access
IEMOCAP	12 hrs, 10 actors	English	Anger, happiness, sadness, frustration, neutral (+ more)	Research (free registration)	USC SAIL Lab — request access
RAVDESS	1 hr, 24 actors	English	Calm, happy, sad, angry, fearful, disgusted, surprised, neutral	CC-BY-NC-SA 4.0	Zenodo / narad/ravdess HuggingFace
CREMA-D	7,442 clips, 91 actors	English	Anger, disgust, fear, happy, neutral, sad	Open (ODC-BY)	m3hrdadfi/crema_d on HuggingFace
EMO-DB	535 utterances, 10 actors	German	Anger, boredom, disgust, fear, happiness, sadness, neutral	CC-BY 4.0	speech31/emodb on HuggingFace
CaFE	~1 hrs, 12 actors	Canadian French	Anger, disgust, fear, happiness, neutral, sadness, surprise	CC BY-NC-SA 4.0	Zenodo
MELD	13,708 utterances	English (TV dialogue)	Anger, disgust, fear, joy, neutral, sadness, surprise	CC-BY-NC 4.0	declare-lab/meld on HuggingFace
MSP-Podcast	237+ hrs, 1,500+ speakers	English (naturalistic)	8 emotions + valence/arousal/dominance	Research (free registration)	MSP Lab UT Dallas — request access

SAVEE	0.8 hrs, 4 male speakers	English	Anger, disgust, fear, happiness, sadness, surprise, neutral	Research (free)	Kaggle
EMOVO	6 actors, ~500 utterances	Italian	Disgust, fear, anger, joy, sadness, surprise, neutral	Research	ISCA Archive
EmoBox	32 datasets, 14 languages	Multilingual	Varies per dataset	Mixed (see per-dataset)	github.com/emo-box/EmoBox — unified toolkit
EmoNet-Voice Big	5,000 hrs synthetic	EN, DE, ES, FR	40 fine-grained emotion categories	CC-BY 4.0	HuggingFace (EmoVoice project)

ASR — Automatic Speech Recognition

Dataset	Size	Languages	License	Access
LibriSpeech	960 hrs EN	English	CC-BY 4.0	openslr/librispeech_asr on HuggingFace
MLS (Multilingual LibriSpeech)	44.5K hrs EN + 6K hrs	EN, DE, FR, ES, IT, PT, NL, PL	CC-BY 4.0	facebook/multilingual_librispeech on HuggingFace
Mozilla Common Voice 18	EN ~3,000 hrs, FR ~900 hrs	100+ languages	CC-0	mozilla-foundation/common_voice_18_0 on HuggingFace
FLEURS	~12 hrs/language	102 languages	CC-BY 4.0	google/fleurs on HuggingFace
VoxPopuli	1,800 hrs labelled	23 European languages	CC-0	facebook/voxpopuli on HuggingFace
GigaSpeech	10,000 hrs EN	English	Apache 2.0	speechcolab/gigaspeech on HuggingFace
TED-LIUM 3	452 hrs	English	CC-BY-NC-ND 3.0	LIUM/tedlium on HuggingFace

AED — Audio Event Detection

Dataset	Size	Classes	Emergency-relevant	License	Access
FSD50K	108 hrs, 51,197 clips	200 AudioSet classes	Sirens, alarms, screaming, crying, glass break, crowd noise	CC-BY (mixed per clip)	philgzi/fsd50k on HuggingFace / Zenodo
UrbanSound8K	8,732 clips ≤4s	10 urban classes	Siren explicitly included, also gun shot, engine	Mixed CC	danavery/urbansound8k on HuggingFace
ESC-50	2,000 clips, 5s each	50 classes	Rain, wind, fire, crying baby, siren, breathing	CC-BY (some NC)	ashraq/esc-50 on HuggingFace / GitHub
DCASE 2024 (TAU Urban Acoustic Scenes)	~64 hrs dev set	10 scene classes	Airport, street-traffic, transit — no sirens	CC-BY 4.0	dcase.community/challenge2024
AudioSet (balanced subset)	~22K clips, 527 classes	527 classes	Broadest emergency coverage — sirens, alarms, explosions, screaming	CC-BY 4.0 (labels only — audio via YouTube)	research.google.com /audioset — ~15-30% links dead

VAD — Voice Activity Detection

Dataset	Size	Content	License	Access
MUSAN	~109 hrs	Music, speech, noise — suitable for VAD training by labelling segments	CC-BY 4.0	openslr.org/17/ / openslr/musan on HuggingFace
LibriParty	LibriSpeech-derived with noise	Multi-speaker scenarios, noise overlaps, silence segments	CC-BY 4.0	SpeechBrain VAD recipe — auto-downloads
LibriVAD	15 GB / 150 GB / 1.5 TB variants	LibriSpeech + diverse noise augmentation, explicit speech/silence labels	Open	arxiv.org/abs/2512.17281 — released 2025
AVA-Speech	Audio from 160 YouTube clips	Clean speech, speech+noise, non-speech — dense VAD labels	Research	research.google.com/ava/ava_speech.html
AMI Meeting Corpus	100 hrs meeting audio	Multi-speaker, naturalistic speech + silence + overlap	CC-BY 4.0	edinburghcstr/ami on HuggingFace
DEMAND	18 environments, multi-channel	Real-world multichannel background noise recordings (offices, streets, transit, cafeterias) — for noise injection when training/testing VAD robustness	CC-BY-SA 3.0	Zenodo

Cross-cutting: Noise Augmentation (used across AED, VAD, ASR, SER fine-tuning)

Dataset	Size	Purpose	License	Access
MUSAN	~109 hrs	Generic noise injection for augmentation	CC-BY 4.0	openslr.org/17
DEMAND	18 environments	Real-world environmental noise injection (18 indoor/outdoor settings) for augmentation	CC-BY-SA 3.0	Zenodo
RIR Noise (OpenSLR 28)	Large	Room impulse responses	Apache 2.0	openslr.org/28

Open sourcely available models

Model	Params	Train Type	Language s	LID	AID	SER	ASR	AED	VAD
facebook/ mms-300m	300M	SSL backbone	1,406	*	*_2	*	*	*_1	*
facebook/ mms-1b	1B	SSL backbone	1,406	*	*_2	*	*	*_1	*
facebook/ mms-1b-all	1B + 2.5M/la ng	Fine-tuned (ASR + adapters)	ASR: 1,162 LID: 4,017	+ (sep. checkp oint)	*_2	*	+	*_1	*
microsoft/ wavlm-larg e	316M	SSL backbone	EN only	-	*_2	*	*	*_1	*
facebook/ wav2vec2-x ls-r-300m	300M	SSL backbone	128 langs	*	*_2	*	*	*_1	*
utter-projec/ mHuBERT- 147	~95M	SSL backbone	147 langs	*	*_2	*	*	*_1	*
FunAudioL LM/ SenseVoice Small	~225M	Fine-tuned	5 langs (ZH, YUE, EN, JA, KO)	+	*_2	+	+	+(parti al)	*
openai/ whisper-lar ge-v3	1.55B	Fine-tuned	99 langs	+	*_2	*	+	*_1	+
openai/ whisper-lar ge-v3-turbo	809M	Fine-tuned	99 langs	+	*_2	*	+	*_1	+
Qwen/ Qwen2-Aud io-7B-Instru ct	7B	Fine-tuned	8+ langs	+	-	+(pro mpt)	+	+(pro mpt)	-
Qwen/Qwe n3-ASR-1.7 B	1.7B	Fine-tuned	52 langs	+	*_2	*	+	-	+
Qwen/Qwe n3-ASR-0.6 B	0.6B	Fine-tuned	52 langs	+	*_2	*	+	-	+

Legend:

- + : Is already available in the model or as an output head
- : Can not be done because of the architecture.
- * : Can be done with an modification
- *_1 : Can be achievable with an addition to the trained backbone with a CNN or LoRA.
- *_2 : Will need extensive dataset which is about accents to make this feature work

Disclaimer:

There are models (such as Jzuluaga/accent-id-commonaccent_xlsr-en-english, Jzuluaga/accent-id-commonaccent_xlsr-de-german) where AID is supported but this would need parallel work of different models. Moreover, there is a model named FunAudioLLM/SenseVoiceLarge but it is only paper available, not publicly available.

Conclusion:

To show all the features in the PoC:

1. Using utter-project/mHuBERT-147, it has the smallest model size which is better since there will be more room to iterate and it can support all the features (LID, AID, SER, ASR, AED, VAD), in theory, but there will be a need for a greatly designed/obtained dataset on SER, AED and VAD.
2. FunAudioLLM/SenseVoiceSmall, viable if language coverage is solved. It ships with LID, SER, ASR, and AED already working (only AID missing), so setup cost is near zero. The blocker is not size (225M is small and fast to iterate on) but languages: it covers only ZH/YUE/EN/JA/KO, none of which match the PoC target. Choosing it means either finding a community fine-tune for our languages or fine-tuning it ourselves on Common Voice / MLS data, which will take a lot of time.
3. Qwen/Qwen3-ASR-0.6B, smallest *capable* model. At 0.6B it's only ~6× mHuBERT's size but comes with ASR, LID, and VAD already working across 52 languages/dialects (Apache 2.0). Only SER, AED, and AID need to be added, versus mHuBERT where all six features start from zero. If iteration speed is the criterion, starting with three features working beats starting with none.
4. Multi-model pipeline, since AID requires separate models anyway (per the disclaimer), a combination minimizes total fine-tuning effort:
 - ASR + LID: Whisper-large-v3-turbo or Qwen3-ASR-0.6B (working out of the box)
 - VAD: Silero-VAD or a LibriVAD-trained model (preprocessing stage)
 - SER: fine-tuned wav2vec2/WavLM head on CaFE + EmoNet-Voice (FR coverage)
 - AED: a classifier trained on FSD50K/UrbanSound8K for siren/alarm/scream classes
 - AID: CommonAccent-style xlsr classifier (as noted in disclaimer)

Tradeoff: more moving parts to orchestrate, but zero features trained from scratch.